

Project 3: Active Learning for Learning-to-Defer

Human-Centric Artificial Intelligence

In this project, you are asked to implement active learning methods for human-AI collaboration through learning-to-defer. For this project, **we are expecting a PDF report**, containing a description of the experiments, a justification of the design choices and a detailed report of the results. The report must be accessible from the project interface (for instance with a download button).

1 Preliminaries: Dataset and Simulated Experts

The goal of this project is to train a classifier that can collaborate with a human expert. Unlike traditional classification, the system is allowed to either make a prediction itself or defer the decision to a human expert. While training such systems, it is important to have access to the expert's predictions, but this data is usually hard to collect. In this project, we propose to collect this data directly from the expert, using active learning.

The task for the project is to train a classifier for the AG News dataset (https://huggingface.co/datasets/fancyzhx/ag_news). The purpose is to label news articles by their topic. The dataset is split into a training set and a test set. The test set will be used only for evaluation.

Your first task consists in training a classification model on the complete labeled training dataset. This classifier will serve as the baseline performance that the human-AI team should aim to match or surpass.

Task 1

Propose a classification model for the chosen dataset. Train the model using all available labels and report the test accuracy.

2 Expert simulation

The idea of a deferral model is to be able to ask an expert for some information. In the context of this project, you will use only a simulated expert. A possible extension is to propose an interactive interface where the user can actually play the role of the expert. Your second task is then to design one or several simulated experts. These experts should not be perfect. Instead, they should exhibit expertise in specific regions of the input space. The expert's predictions will be available when queried.

Task 2

Design and implement at least one simulated expert. Analyze its strengths and weaknesses on the dataset and report its accuracy on the test set.

3 Learning to Defer

We now consider a setting in which both the classifier and the expert labels are available.

Task 3

Implement a learning-to-defer strategy. Given an input, the system must choose between producing a prediction itself or querying the expert. Report the performances of the trained system. The evaluation should reflect not only classification accuracy but also the quality of the deferral decisions.

4 Active Learning for expert competence discovery

From now on, we assume that we do not have access to any expert label during the training. This makes it impossible to use learning-to-defer methods, since we cannot learn when deferral is actually useful. The objective is then to actively select which examples should be presented to the expert in order to efficiently learn both the classification task and the expert's competence profile. At this stage, the algorithm:

- Has access to the whole training dataset to learn the classifier
- Does not have access to any expert data
- Can query the expert when relevant to get a new label on a specific data point

Task 4

Choose an active learning strategy for querying the expert. The goal is to efficiently learn when deferral is beneficial. Justify your choice and report the results using metrics of your choice.

5 (Optional) Active learning with human expert

Task 5

This task is optional, only a suggested extension of the project.

Implement an interface where a user can interact with the system and provide the requested labels.